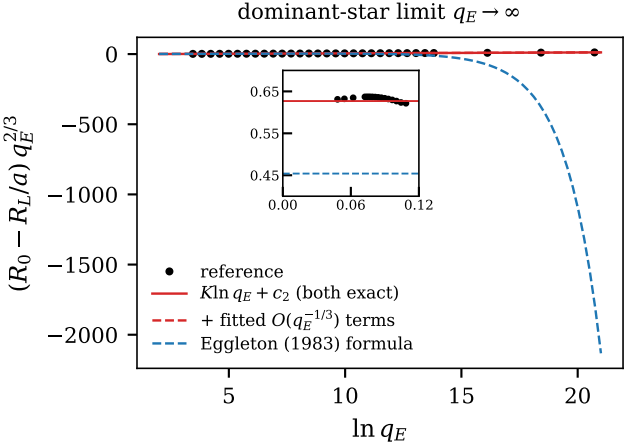
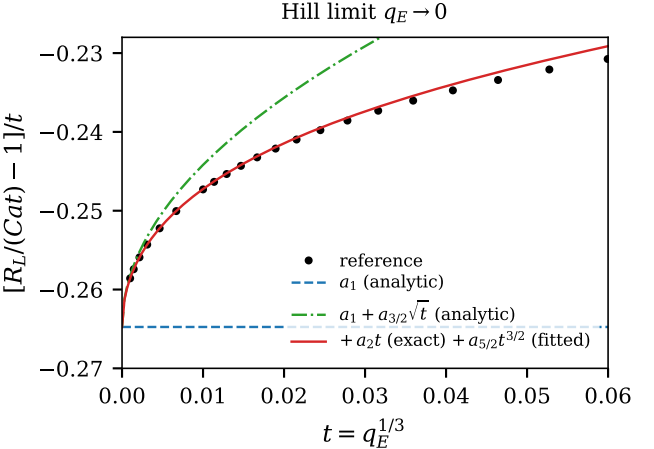

dominant-star limit $q_E \to \infty$
$(R_0 - R_L/a)\, q_E^{2/3}$
0
−500
−1000
−1500
−2000
5
10
15
20
$\ln q_E$
0.65
0.55
0.45
0.00
0.06
0.12
reference
$K \ln q_E + c_2$ (both exact)
+ fitted $O(q_E^{-1/3})$ terms
Eggleton (1983) formula


Hill limit $q_E \to 0$
$[R_L/(Cat) - 1]/t$
−0.23
−0.24
−0.25
−0.26
−0.27
0.00
0.01
0.02
0.03
0.04
0.05
0.06
$t = q_E^{1/3}$
reference
a_1 (analytic)
$a_1 + a_{3/2}\sqrt{t}$ (analytic)
$+ a_2 t$ (exact) $+ a_{5/2} t^{3/2}$ (fitted)